

# Sachin Avutu

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## Education

### Drexel University

*Bachelor of Science in Computer Science, Minor in Linguistics*

*Sep 2024 – Expected Jun 2029*

- **GPA:** 3.04
- **Relevant Coursework:** Advanced Programming Tools and Techniques, Computing Informatics and Design I – III, Computer Programming I – II, Calculus I – II, Linear Algebra, Linguistics 101, Principles of Microeconomics

## Experience

### Full-stack Developer

*The Triangle - Drexel University Student Newspaper*

*Philadelphia, PA*

*May 2025 – Present*

- Led migration from WordPress to a custom headless CMS in Go, delivering scalable content management via a REST API serving a React frontend
- Architected a Python ETL pipeline migrating 11+ years of WordPress data (18k+ articles, 5k+ authors) into MariaDB, preserving data continuity despite upstream failures
- Developed a MinHash-based entity resolution system using NumPy and Numba to reduce duplicate records and improve dataset integrity in production
- Maintained a fully self-hosted Proxmox cluster, ensuring 99.25% site uptime

### Software Engineer Co-op

*eMoney Advisor*

*Radnor, PA*

*Sep 2025 – Mar 2026*

- Delivered production features and reduced technical debt in a 1.3M+ LOC .NET codebase supporting integrations with 12,000+ financial institutions
- Traced and corrected a partner API change that caused a spike in client-facing errors using Elasticsearch and Kibana
- Developed and deployed eMoney's Rocket Mortgage integration to production alongside senior engineers
- Stabilized production releases with TDD-driven NUnit test suites and Jenkins/Octopus CI/CD deployment pipelines

## Projects

### TriApply ATS | [github.com/ssavutu/TriApply-ATS](https://github.com/ssavutu/TriApply-ATS)

*Jun 2026 – Present*

- Developing an applicant tracking system in Clojure and ClojureScript, with a Google Forms style application and Reagent/React reviewer portal
- Designing a data-only EDN DSL that drives both form rendering and server-side validation
- Implementing a Ring/Jetty and reitit API with multipart submission, UUID-keyed idempotency, and secure upload storage
- Integrating Apache Tika for resume text extraction, enabling structured parsing and future matching against role requirements

### LittleBrother | [github.com/ssavutu/LittleBrother](https://github.com/ssavutu/LittleBrother)

*Jul 2025 – Present*

- Engineering an HNSW approximate-nearest-neighbor vector index from scratch in C++, implementing graph-based search for dense embeddings
- Implementing probabilistic level assignment and greedy top-down layer descent for logarithmic-time insertion
- Writing cache-friendly L2-squared and cosine distance kernels over dense float embeddings
- Building a Python embedding pipeline using SigLIP2 to generate real vector inputs for indexing

### Minhashlib | [github.com/ssavutu/minhashlib](https://github.com/ssavutu/minhashlib)

*Feb 2026 – Jun 2026*

- Released an MIT-licensed Python library for MinHash-based similarity detection on large string datasets
- Achieved a 6× CPU speedup over the datasketch library while maintaining comparable accuracy
- Accelerated signature generation through vectorization and JIT compilation using Numba
- Integrated the library into a production ETL pipeline at The Triangle

### EagleEyePhilly | [github.com/ssavutu/EagleEyePhilly](https://github.com/ssavutu/EagleEyePhilly)

*May 2026*

- Built a FastAPI service backed by Selenium automation during a hackathon to expose Philadelphia 311 workflows via a programmable API
- Deployed a demo-ready backend via Cloudflare Tunnel from a local machine for live end-to-end demos
- Partnered on Gemini integration to support one-shot situational analysis and resolved cross-team backend integration issues

## Skills

**Languages:** Python, Go, C#, C++, Clojure, ClojureScript, SQL, JavaScript, TypeScript, HTML, CSS

**Technologies:** .NET, FastAPI, Ring, reitit, React, NUnit, Selenium, MariaDB, NumPy, Numba, Transformers, Gemini API

**Tools:** Git, GitHub, Linux, Docker, Proxmox, Jenkins, Octopus, Jira, Confluence, Elasticsearch, Kibana, Bitbucket, Cloudflare